

Course : Energy Systems Modeling

Professor : Dr. Khajepour

Exercise Topic : GAMS

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Given the demo version limitations, we must find an alternative: either reducing equations or solving in intervals (e.g., every two years) to plot and predict trends. Since we can run up to 7 years without hitting the limit, I will not reduce the equations to maintain quality; instead, I will use forecasting and run the code in multiple stages.

To forecast the 10-year and 15-year trends, we first execute the demo version at its maximum capacity for the first seven years and analyze it with the code and output available in the exercise ZIP file.

The model achieved an objective value of 963,926.7 over a 7-year horizon, proving its complete stability in large-scale scenarios.

PV and CHP Capacity Analysis:

The negative value for Cap(PV) indicates a strong tendency for more installation, which we have capped. This technology has proven profitable over this horizon. For Cap(CHP), the output remains at the lower bound of 200. Although its initial cost is high, it is a strategic choice. Unlike solar panels, which are weather-dependent, CHP ensures system reliability during outages or transmission failures.

Efficiency and Industrial Reality:

The output shows realistic efficiency levels. The CHP electrical efficiency is around 35% (2.85 units of gas per unit of electricity), which aligns with national power plant standards. The Boiler operates at 90% efficiency. In this 7-year model, the system prioritizes CHP for heat, using the Boiler only to cover final deficits—a shift from the 2-year model's behavior.

Economic Decisions (Electric Water Heater):

The capacity for the ElecWaterHeater is zero. Even though it has a low initial cost, the high price of electricity (three times that of gas) makes it uneconomical in the long run. The model has correctly discarded it to minimize the total objective function value, which stands at 963,926.7.

Grid Dependency and Self-Sufficiency:

The results show that CHP covers the base load. In winter nights, the system becomes fully self-sufficient, requiring no grid power. However, during summer days, we still rely on the grid, though not entirely. Electricity is more critical and expensive during the first half of the year. The rising “Shadow Price” for heat in later years suggests that meeting winter heat demand will become increasingly difficult and costly.

Why CHP Dominates:

Despite high operational and initial costs, CHP is the winner for several reasons:

Solar Limitations: While PV is cheaper, its maximum capacity is limited.

Grid Costs: The annual grid subscription fee (788\$/year kW) is nearly five times higher than CHP's operating costs.

Cogeneration: CHP provides free heat while generating power, leading to high overall efficiency and 24 hours availability.

Analysis of Marginal Values and Seasonal Strategy:

A key observation in the output is the Marginal Value (shadow price), which represents how much the objective function changes with a one-unit shift in a constraint. In the final three years, the marginal value for CHP becomes negative. This indicates that the CHP unit has reached its maximum electrical capacity, making this a "binding" or "active" constraint.

Why does CHP hit its limit in Winter instead of Summer?

One might expect the peak to occur in summer due to cooling demands, but the model chooses winter for several logical reasons:

Thermal Efficiency and Heat Waste: CHP produces electricity and heat simultaneously. In summer, hitting maximum electrical output would generate excess heat that isn't needed (where the shadow price of heat is likely zero). Dumping this excess heat is inefficient both economically and environmentally.

Winter Demand Sync: In winter, every unit of electricity produced by the CHP comes with "free" heat that is fully utilized by the system. This minimizes waste and maximizes the total return on investment.

The Role of PV: In the summer, Solar PV acts as a strong competitor to CHP. It provides free electricity during the day, reducing the system's reliance on CHP. In the winter, however, PV generation drops, and the CHP must step up to cover both power and heating loads.

Conclusion:

Your optimized system for the 7-year horizon has these features: PV has reached its maximum limit and still has value for further expansion. CHP is the backbone of the system, providing both power and heat. The Boiler only acts as a small backup at the end of the period. The Electric Heater is completely uneconomical. The power grid is still necessary, but reliance on it has decreased. The main bottlenecks in the final years are winter days, CHP power and heat capacity limits, and the small boiler capacity

Now we will run the model for a 7 to 10 year horizon and analyze its output :

Model analysis for 10 years:

In analyzing the 7 to 10-year horizon, we examine the marginal values of the technologies. The PV value (-204.77) indicates that PV remains highly economical; if the cap were not 100, the model would install more capacity. As noted previously, the negative value means that every 1-unit increase in the PV upper bound reduces the total cost by 204.77 units. Similarly, EWH (+20.11) is positive, meaning that if we are forced to install 1 unit, the cost increases by 20.11 units, making it completely uneconomical. For CHP, this value has slightly decreased compared to the first 7 years but remains significant, while for the boiler, it is almost constant.

The electricity production trend during these years is as follows: in the first half of the year, from year 7 to 10, the trend is upward during the day, nearly constant at night, constant during the second half of the day, and upward at night. In the second half of the year during the day, we have reached the CHP electrical production capacity limit. Regarding heat production, there is an increase from year 7 to 8, reaching the production capacity limit, and then this production remains constant until year 10. The boiler, which had no significant role in previous years, becomes effective in year 8 and reaches its production capacity limit in year 10. During cold season nights, grid electricity purchases are zero, and this has not changed up to year 10. During the second half of the year, heating demand is higher during the day than at night. Since PV and the boiler have reached their capacity limits, we are forced to use the Electric Water Heater for heating; this technology requires electricity, which is not fully supplied by the CHP, so we buy from the grid, and this amount has increased every year. In the first half of the year, grid electricity purchases have grown steadily to meet the building's demand (we do not use CHP to meet the full demand due to heat waste). Gas consumption growth is logical and moderate because its main consumer is the CHP, which has remained somewhat constant and managed the load growth.

In nime2.day, the Marginal value grew significantly from y7 to y10:

0 , 56.8 , 58.4 , 98.7

This means the shadow cost of providing heat during winter days is increasing year by year and peaks in y10. This is the main bottleneck of the system.

Regarding CHP, from year 8 onwards in the second half of the year, the heat capacity constraint has become active, reaching its most severe state in year 10. In the summer (nime1), the CHP's minimum heat constraint is always active. This means that to produce electricity, the CHP is forced to generate extra heat that has no use during the summer.

The constraints for year 10 include:

CHP_MaxElec is active (marginal = -135.86)

CHP_UpperHeat is active (marginal = -98.73)

Boiler_limit is active (marginal = -40.33)

This means that any increase in demand will require an increase in capacity.

10-Year Overall Summary:

- PV reaches its maximum capacity (100) and would be profitable to expand if the limit increased.

- CHP is the main backbone for electricity and heat. It stops around 198 (even though 700 was allowed) because higher capacity would cause heat dumping and becomes inefficient.
- Boiler is small in early years, but becomes important from Year 8 and reaches full use by Year 10 to cover remaining heat demand.
- Electric Water Heater (EWH) is mostly uneconomical due to high electricity cost, but is used as a last resort in the final years when heat is still short.
- The main problem area is winter heat supply. By Year 10 the heat shadow price becomes very high, showing serious heat shortage.
- Grid electricity is still needed, especially in summer. Winter nights can become self-sufficient, but summer days still require grid purchases.
- By Year 10, the system is limited by: maximum CHP electrical output, maximum CHP thermal output, and boiler capacity. Any further demand growth after Year 10 will mostly be met by expensive grid imports.

Next we will run the project for the 10th to 15th year.

Model analysis for 15 years

As predicted, the final outputs for these years are:

PV = 100.000

CHP = 198.135

Boiler= 4.360

PV remains at its limit of 100. This means that PV is economically very attractive, but the capacity cap has prevented further growth. CHP has reached 198.135. This figure shows that CHP remains the main source of energy supply and operates at near-saturation in the long-term horizon. The Boiler equals 4.360. Its role remains small and supplementary, used only for covering heating peaks.

An interesting point in this output is the Marginal values of the HeatBalance equation during the cold season (nime2.day):

Years 10 to 12: The Marginal value is EPS (nearly zero). This means the system supplied heat with difficulty but without extra cost.

Year 13: The Marginal value suddenly reaches 23.6.

Year 14: It jumps to 58.4.

Year 15: It reaches 98.7

Analysis: This indicates that from year 13 onwards, the system faces a severe heat shortage. Each additional unit of heat in year 15 costs the system 98 units. This shows that demand growth in the last 5 years has pushed the system to its thermal capacity limit.

Look at the Marginal values for CHP_MaxElec and CHP_UpperHeat in years 13 to 15 (large negative values such as -135.8 and -98.7):

These numbers mean that during these years, the CHP is locked exactly at its maximum power capacity.

Also, by observing the growth in demand and the trend of electricity purchases from the grid, we realize that since the PV capacity is fixed at 100 and the CHP has reached its limit in the final years, all building load growth is directly supplied through grid purchases.

An interesting finding is that the Gas Marginal in summer (78.8) is higher than in winter (52.5). This is likely because in summer, the CHP operates solely for electricity and its heat is dumped; therefore, the "value" of each unit of gas consumed for electricity generation effectively costs the model more in the summer.

For CHP, the value has remained constant at 198 during these years. The model behaves as if this is the technology's limit. Although a capacity of 700 kW was provided in the problem description, my research indicates that 700 kW represents the total combined thermal and electrical power. The model did not increase the value beyond 198 because it is the technology's economic optimum; furthermore, increasing it would ultimately result in nothing but wasted heat for the system.

Another interesting observation in the output is the 0.247 coefficient in the PV_limit constraint for Cap(PV):

This means the model assumes that every 1 kW of solar panels produces 0.247 units of electricity during daylight hours (day). Furthermore, in the second half of the year (nime2), this number drops to 0.114. This reflects the decrease in solar radiation during winter, which forces the model to put more pressure on the CHP and the grid during the cold season.

The following efficiencies are observed in the output:

CHP Electrical Efficiency: A coefficient of -2.857 in the Gas_balance. Calculation: $1/2.857=0.351$ / $2.857 = 0.351/2.857=0.35$. This means the CHP electrical efficiency is set at 35%.

Boiler Efficiency: A coefficient of -1.111 in the Gas_balance. Calculation: $1/1.111=0.901$ / $1.111 = 0.901/1.111=0.90$. This means the boiler efficiency is 90%.

Heat-to-Power Ratio: A coefficient of 1.3 in CHP_UpperHeat. This means that for every 1 unit of electricity, the CHP can produce a maximum of 1.3 units of heat.

INFES:

The term INFES appears in front of some equations (for example, 233.3 for ElecBalance).

This means that at the starting point (when all variables are zero), the system has a power deficit of 233 units.

15-Year Overall Summary:

- PV has reached its limit.

- Despite the limit of 700, CHP stopped at 198.135. This means its own technical and economic limits were the deciding factors, rather than the allowed limit of 700.
- The Boiler only operated at a minimal level.
- The objective value of 832,251.5403 shows that the model reached a final solution, but it was recorded in the report with a NONOPT label.

Third question

Set CHP capacity to zero, run the simulation, and find the difference between the Z values.

In the optimal scenario, $Z = 963,926$. When the CHP is removed, Z increases to 1,116,800, creating a difference of 152,873. Note that due to Demo version limitations, this run covers a 7-year period; for a 10-year horizon, this gap would be even wider. Removing this technology results in a 16% increase in total costs. In this scenario, grid electricity purchases increase fourfold (a 256% rise), and the system is forced to utilize Electric Water Heaters (EWH) and expand boiler capacity to meet the thermal demand.

Question 4:

First, we run the code from year $A+2$, which is equal to Year 6, up to Year 10. To answer this question, we use the model output to calculate the total electricity and gas demand. According to the HeatBalance and ElecBalance equations, the right-hand-side values represent the demand in each time period. In the new exercise, cooling does not have a separate equation and is considered as part of electricity consumption.

Then, we define a condition: this technology will be selected in the system only if its present value over the considered period is less than or at most equal to the total savings in electricity and gas. Therefore, the answer to this question is the sum of the 20% savings over the period from Year 6 to Year 10.

$NPV(\text{technology}) \leq PV \text{ of savings from reduced heating + cooling demand}$

```

17 ---- EQU ElecBalance
18
19 LOWER LEVEL UPPER MARGINAL
20
21 nime1.day .y6 233.3333 233.3333 233.3333 236.5200
22 nime1.day .y7 235.6667 235.6667 235.6667 236.5200
23 nime1.day .y8 238.0233 238.0233 238.0233 236.5200
24 nime1.day .y9 240.4036 240.4036 240.4036 236.5200
25 nime1.day .y10 242.8076 242.8076 242.8076 236.5200
26 nime1.night.y6 140.0000 140.0000 140.0000 236.5200
27 nime1.night.y7 141.4000 141.4000 141.4000 236.5200
28 nime1.night.y8 142.8140 142.8140 142.8140 236.5200
29 nime1.night.y9 144.2421 144.2421 144.2421 236.5200
30 nime1.night.y10 145.6846 145.6846 145.6846 236.5200
31 nime2.day .y6 180.0000 180.0000 180.0000 157.6800
32 nime2.day .y7 181.8000 181.8000 181.8000 157.6800
33 nime2.day .y8 183.6180 183.6180 183.6180 157.6800
34 nime2.day .y9 185.4542 185.4542 185.4542 157.6800
35 nime2.day .y10 187.3087 187.3087 187.3087 157.6800
36 nime2.night.y6 120.0000 120.0000 120.0000 150.1714
37 nime2.night.y7 121.2000 121.2000 121.2000 150.1714
38 nime2.night.y8 122.4120 122.4120 122.4120 150.1714
39 nime2.night.y9 123.6361 123.6361 123.6361 150.1714
40 nime2.night.y10 124.8725 124.8725 124.8725 150.1714
41
42 ---- EQU HeatBalance
43
44 LOWER LEVEL UPPER MARGINAL
45
46 nime1.day .y6 20.0000 20.0000 20.0000 -28.1571
47 nime1.day .y7 20.2000 20.2000 20.2000 -28.1571
48 nime1.day .y8 20.4020 20.4020 20.4020 -28.1571
49 nime1.day .y9 20.6060 20.6060 20.6060 -28.1571
50 nime1.day .y10 20.8121 20.8121 20.8121 -28.1571
51 nime1.night.y6 - - - -28.1571
52 nime1.night.y7 - - - -28.1571
53 nime1.night.y8 - - - -28.1571
54 nime1.night.y9 - - - -28.1571
55 nime1.night.y10 - - - -28.1571
56 nime2.day .y6 200.0000 200.0000 200.0000 EPS
57 nime2.day .y7 202.0000 202.0000 202.0000 EPS
58 nime2.day .y8 204.0200 204.0200 204.0200 40.2123
59 nime2.day .y9 206.0602 206.0602 206.0602 58.4000
60 nime2.day .y10 208.1208 208.1208 208.1208 98.7346
61 nime2.night.y6 140.0000 140.0000 140.0000 EPS
62 nime2.night.y7 141.4000 141.4000 141.4000 EPS
63 nime2.night.y8 142.8140 142.8140 142.8140 EPS
64 nime2.night.y9 144.2421 144.2421 144.2421 EPS
65 nime2.night.y10 145.6846 145.6846 145.6846 EPS
66

```

تقاضای سرمایش (کیلووات ساعت)		تقاضای گرمایش (کیلووات ساعت)	
شب	روز	شب	روز
۶۰	۱۶۰	۰	۲۰
۰	۰	۱۴۰	۲۰۰

Total heating demand=1836.4

Total cooling demand= 1122.2

$NPV = 0.20 \times (\text{Total heating demand} \times \text{PriceGas} + \text{Total cooling demand} \times \text{PriceElec})$

Y6= TotalSaving = **10294.2516**

Y7= TotalSaving = **10397.194116**

Y8= TotalSaving = **12141.98874636**

Y9= TotalSaving = **13012.96096636**

Y10=TotalSaving = **14821.9841713428**

Total=NPV=35014.0262

That is, if the NPV of the technology is less than or equal to this value, we use it.

The fifth question:

To solve this question we slightly modify the structure of the code and add the carbon tax information. We also define two outputs so that we can compare the results. In fact, the code is run twice, and finally, the outputs are printed.


```

----      202 PARAMETER z_1                = 963926.705
          PARAMETER z_2taxed              = 1493611.114
          PARAMETER ExtraCost              = 529684.409

----      202 PARAMETER Cap_1

PV      100.000,    CHP    200.116,    Boiler    4.403

----      202 PARAMETER Cap_2taxed

PV      100.000,    CHP    200.116,    Boiler    4.403

----      202 PARAMETER Cap_change

          ( ALL      0.000 )

```

As can be seen, the total system cost has increased by 529684.41 which is equivalent to **56%**

We have four technologies, one of which is PV, which is both cost-effective and fuel-free. Among the other three technologies, two operate on natural gas and one on electricity, namely ElecWaterHeater_. This technology played a negligible role in the building previously, and now, given that the carbon tax on electricity use is three times higher than on gas, it has been completely phased out.

In the previous analysis, we established that CHP is a cogeneration system; it produces electricity and heat simultaneously. Using this energy is fine if it meets the demand, but any surplus is dumped because we lack storage capacity. With the current carbon tax, this surplus production leads to greater losses.

If carbon tax were applied to all energy consumption, then—for example—when we need electricity, purchasing it from the grid would incur a tax on the electricity used for cooling. Conversely, if we generated that electricity via CHP, we would have to pay the tax on the gas consumed, and then pay an additional tax on the usage of that generated electricity. The model correctly identifies that under such conditions, CHP is sometimes economically unjustifiable, leading to a reduction in its share.

However, since the tax is currently applied only to grid purchases, we must compare CHP directly against the grid. Given that the CHP has an electrical efficiency of **35%**, it requires 2.85 units of gas to produce 1 unit of electricity. Since the carbon tax on electricity is 2.6 times higher than that on gas a ratio lower than 2.85 grid electricity technically carries a lower tax burden. However, this difference is negligible and does not significantly impact the model. When combined with the high cost of grid electricity, using CHP remains the more economical choice.

Furthermore, we must account for the heating challenges in winter; even if grid electricity were cheaper, we are forced to operate the CHP at full capacity to meet thermal demand, unless the boiler capacity were increased. Consequently, while the technology mix remains unchanged, our total system costs increase significantly. In this scenario, it is essential to prioritize the expansion of PV and other renewable energy sources.

The sixth question:

When we run the code under the constraint that the model must achieve a 30% reduction in emissions, the solver cannot find an optimal solution. Since the previous case was already in a poor condition and operating at the edge of feasibility (i.e., just able to meet the demands), the model is unable to simultaneously satisfy the energy requirements and reduce emissions by 30%

```

3550 ---- VAR =          -INF      8028679.7331      +INF      .
3551
3552
3553 **** REPORT SUMMARY :      0      NONOPT
3554                          30 INFEASIBLE (INFES)
3555                          SUM      217435.9861
3556                          MAX      34391.8931
3557                          MEAN      7247.8662
3558                          0 UNBOUNDED
3559 GAMS 25.1.3 r4e34d435fbd Released Oct 30, 2018 WEX-WEI x86 64bit/MS Windows - 05/29/26 17:42:12 Pa
3560 General Algebraic Modeling System
3561 Execution
3562
3563
3564 ---- 219 PARAMETER z_3          =      8028679.733
3565
3566
3567 EXECUTION TIME      =      0.000 SECONDS      3 MB 25.1.3 r4e34d435fbd WEX-WEI
3568
3569
3570 USER: CS 635 Spring 2014          G140121/0001AO-GEN
3571 Prof. Michael Ferris, University of Wisconsin-Madison DC2937
3572 License for teaching and research at degree granting institutions
3573 |
3574
3575 **** FILE SUMMARY
3576

```

It is so unstable that it can only achieve emission reductions of a few tenths of a percent.

```

          LOWER          LEVEL          UPPER          MARGIN
--- VAR z          -INF          977830.9624          +INF          .

*** REPORT SUMMARY :          0          NONOPT
                              0 INFEASIBLE
                              0 UNBOUNDED

GAMS 25.1.3 r4e34d435fbd Released Oct 30, 2018 WEX-WEI x86 64bit/MS Windows
e n e r a l   A l g e b r a i c   M o d e l i n g   S y s t e m
x e c u t i o n

---      219 PARAMETER z_3          =          977830.962

EXECUTION TIME          =          0.000 SECONDS          3 MB  25.1.3 r4e34d435fbd WE

SER: CS 635 Spring 2014          G140121/0001AO-GEN
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*** FILE SUMMARY

input      C:\Users\mahdi\Desktop\modelasli.gms

```

In the case where we become exempt from the carbon tax by reducing emissions, the maximum achievable emission reduction is **up to 3%**, and this would cost **\$13,904**

Sources:

Professor Khajepour slides, the TA explanations, Google searches, and the use of Gemini to resolve some unclear errors as well as to correct certain writing issues in the text.